Project 3

Medical Calculations

Investigate how an LLM solves medical calculation questions and what a reported correct number actually demonstrates. Build human evaluation evidence, examine final-answer and displayed-calculation errors, and test whether an LLM judge can apply the group's rubric reliably.

Data and starting materials

The package contains 160 questions across 16 calculators: 32 development and 128 test questions, plus 16 separate prompting examples. Calculator references specify formulas, scoring rules, units, and conventions. The notebook provides model setup, the original answer metric, annotation exports, agreement calculations, prompt helpers, and a judge interface.

Guided warm up

Load the data in Colab and run or inspect saved direct answers. Independently calculate selected cases using the calculator references before consulting model outputs or patient-specific answers. Compare human judgments, study the original tolerance and rounding policy, and preserve development and reserved judge-validation cases.

Required group exploration

1. **Examine answer and process evaluation.** Implement and justify a revised final-answer policy on saved development outputs. Then compare direct answering, zero-shot worked calculations, and one-shot worked calculations on matched questions. Inspect inputs, formulas, units, arithmetic, and formatting; a correct final number does not establish a valid displayed calculation.
2. **Build and validate an LLM judge.** Translate the human rubric into a judge, examine development errors, and make one meaningful improvement. Freeze the rubric and judge before reserved validation. Report agreement, missed errors, false passes, unsupported judgments, and call or parsing failures by response type and dimension.
3. **Analyze solution failures.** Locate the earliest supported error and distinguish its downstream effects from separate mistakes. Compare patterns across methods and calculator categories, including correct-answer/wrong-process cases. Use the settled judge together with human evidence and identify conclusions relying only on automatic judging.

Individual open questions

1. Which evaluation dimensions can your judge handle credibly, and which still need human judgment? Support the boundary with observed cases and validation results.
2. What targeted data, prompting, or evaluator improvement could address an observed failure while preventing leakage? Explore a bounded follow-up or develop a testable design grounded in your results.

Scope

Use one solving model and a reproducible subset if needed. Keep evaluated patient gold answers out of solver prompts, separate tuning from final evaluation by question, and retain unsuccessful outputs in coverage accounting. Paid APIs and model fine-tuning are optional.